

Analysis of arthropod communities in Hawaii using METE Theory

This supplement walks through the steps of recreating the entire analysis presented in the main text. It also provided further analysis in support of and to contextualize results in the main text.

Computing setup

Analyses are supported by a custom R package *HawaiiArthMETE* [foo] which can be installed from github.

```
devtools::install_github("ecoevomatics/HawaiiArthMETE")
```

The quarto [quarto] document used to generate this supplement can be found with the R command `system.file("ms/hawaii-arth-mete-supp.qmd", package = "HawaiiArthMETE")`.

The following R computing environment is used for all analyses:

```
library(meteR)
library(ggplot2)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
library(vegan)
```

Loading required package: permute

```
library(metafor)
```

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Loading required package: metadat

Loading required package: numDeriv

Loading the 'metafor' package (version 5.0-1). For an introduction to the package please type: help(metafor)

Attaching package: 'metafor'

The following object is masked from 'package:vegan':

permutest

```
library(HawaiiArthMETE)
```

Attaching package: 'HawaiiArthMETE'

The following object is masked from 'package:meterR':

arth

```
knitr::opts_chunk$set(  
  # default figure dims  
  fig.width = 5.5, fig.height = 4,  
  # caching is default for all subsequent chunks  
  cache = TRUE  
)  
  
# load data  
data(arth)  
  
sessionInfo() |>  
  print(locale = FALSE)
```

R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.0.1

Matrix products: default

BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework

LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] HawaiiArthMETE_0.1.0 metafor_5.0-1 numDeriv_2016.8-1.1
[4] metadat_1.6-0 Matrix_1.7-4 vegan_2.7-2
[7] permute_0.9-8 tidyr_1.3.1 dplyr_1.2.1
[10] ggplot2_4.0.1 meter_1.2

loaded via a namespace (and not attached):

[1] gtable_0.3.6 jsonlite_2.0.0 compiler_4.5.2 tidyselect_1.2.1
[5] parallel_4.5.2 cluster_2.1.8.1 splines_4.5.2 scales_1.4.0
[9] yaml_2.3.10 fastmap_1.2.0 lattice_0.22-7 R6_2.6.1
[13] generics_0.1.4 knitr_1.50 MASS_7.3-65 tibble_3.3.0

[17] pillar_1.11.0	RColorBrewer_1.1-3	rlang_1.2.0	mathjaxr_2.0-0
[21] xfun_0.52	S7_0.2.1	cli_3.6.5	mgcv_1.9-3
[25] withr_3.0.2	magrittr_2.0.4	digest_0.6.39	grid_4.5.2
[29] rstudioapi_0.17.1	nlme_3.1-168	lifecycle_1.0.5	vctrs_0.7.3
[33] evaluate_1.0.4	glue_1.8.0	farver_2.1.2	rmarkdown_2.29
[37] purrr_1.1.0	tools_4.5.2	pkgconfig_2.0.3	htmltools_0.5.8.1

METE calculations

Comparison of the observed data to predictions from the Maximum Entropy Theory of Ecology (METE) is performed with package *meteR* [rominger-merow2016]. The following code records information about each sample (name, flow age, proportion non-native presence) as well as several metrics of deviation between theory and observation:

- a z^2 value measuring a normalized difference in likelihoods between observed data and theoretical expectation
- differences in proportion of rarity observed and expected
- differences in proportion of dominance observed and expected

All metrics are calculated for both the species abundance distribution (SAD) and individual metabolic rate (or power) distribution (IPD).

```
arth_mete <- split(arth, arth[, c("site", "tree")], drop = TRUE) |>
  lapply(function(x) {
    # the esf for this grouping
    esf <- meteESF(spp = x$species_code,
                   abund = x$abundance,
                   power = x$metabolic_rate)

    # sad and ipd
    s <- sad(esf)
    p <- ipd(esf)

    data.frame(
      # sample info
      site = x$site[1],
      site_name = x$site_name[1],
      tree = x$tree[1],
      flow_age_my = x$flow_age_my[1],

      # z2 values
      z2_sad = logLikZ(s)$z,
```

```

    z2_ipd = logLikZ(p)$z,

    # prop non-native for abundance and power
    prop_n_non_nat = sum(
      x$abundance[!(x$origin %in% c("endemic", "indigenous"))]
    ) /
      sum(x$abundance),
    prop_p_non_nat = sum(
      x$metabolic_rate[!(x$origin %in% c("endemic", "indigenous"))]
    ) /
      sum(x$metabolic_rate),

    # differences in extreme values
    n1_diff = mean(meteDist2Rank(s) == 1) - mean(s$data == 1),
    nmax_diff = sum(meteDist2Rank(s)[1:3]) - sum(s$data[1:3]),
    p1_diff = mean(meteDist2Rank(p) <= 2) - mean(p$data <= 2),
    pmax_diff = sum(meteDist2Rank(p)[1:3]) - sum(p$data[1:3])
  )
})

arth_mete <- do.call(rbind, arth_mete)

```

An initial investigation of patterns in the SAD reveal two clear outliers in terms of deviation from theory and proportion of non-native individuals (Figure 1).

Investigating highly invaded, highly dominated samples

The two outlier samples both show a difference (theory - observed) in maximum abundance of over -300 (Figure 1).

```

outliers <- filter(arth_mete, nmax_diff < -300)

```

```

pivot_longer(outliers,
  c(prop_n_non_nat, nmax_diff),
  names_to = "metric")

```

```

# A tibble: 4 x 12

```

	site	site_name	tree	flow_age_my	z2_sad	z2_ipd	prop_p_non_nat	n1_diff	p1_diff
	<fct>	<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	LA	Laupāhoē	2	0.005	8.33	26.4	0.395	-0.157	0.180

```

2 LA    Laupāhoe~    2      0.005   8.33 26.4      0.395 -0.157 0.180
3 KH    Kohala      6      0.15    6.77 0.300      0.575 -0.186 -
0.0392
4 KH    Kohala      6      0.15    6.77 0.300      0.575 -0.186 -
0.0392
# i 3 more variables: pmax_diff <dbl>, metric <chr>, value <dbl>

```

```

select(arth_mete, site_name, prop_n_non_nat, nmax_diff, z2_sad) |>
  # pivot so both metrics are plotted in faceted panels
  pivot_longer(prop_n_non_nat:nmax_diff, names_to = "metric") |>
  ggplot(aes(value, z2_sad, color = site_name)) +
  geom_point() +

  # add arrows pointing to outliers
  geom_segment(
    data = pivot_longer(outliers,
                        c(prop_n_non_nat, nmax_diff),
                        names_to = "metric"),
    aes(x = value * 0.8, y = z2_sad * 0.9,
        xend = value * 0.97, yend = z2_sad * 0.99),
    arrow = arrow(length = unit(1, "char")),
    inherit.aes = FALSE
  ) +

  # facet by metric and trick strip labels into being
  # axis labels
  facet_grid(
    cols = vars(metric), scales = "free_x",
    labeller = as_labeller(
      c(nmax_diff = "Diff. in maximum abundance",
        prop_n_non_nat = "Prop. non-native individuals")
    ),
    switch = "x"
  ) +
  ylab(expression("SAD"~z^2*"-value")) +
  this_ms_look() +
  theme(strip.background = element_blank(),
        strip.placement = "outside",
        panel.border = element_rect(colour = "black",
                                     fill = NA,
                                     linewidth = 1),
        axis.title.x = element_blank())

```

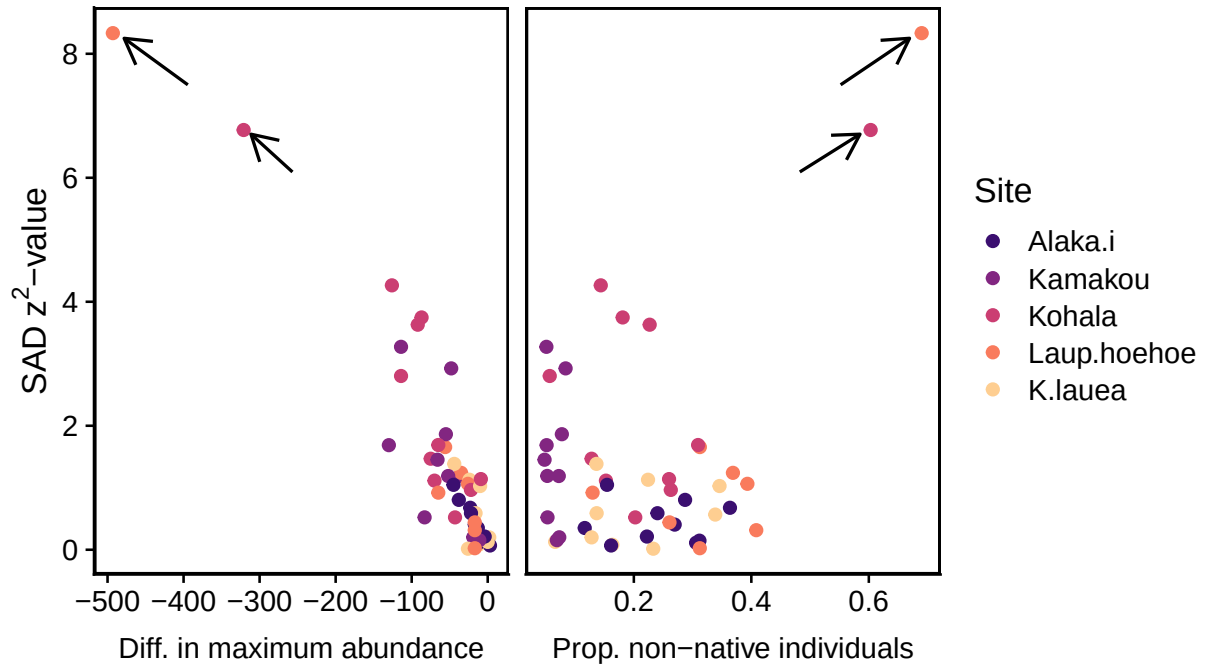


Figure 1

These two outlier samples come from the Laupāhoehoe and Kohala sites. To further understand these two samples and determine if they should remain in the analyses we look at their composition. Table 1 shows the three most abundant species for the Laupāhoehoe outlier sample while Table 2 shows the three most abundant species for the 'r Kohala outlier sample.

```
filter(arth, tree == outliers$tree[1] & site == outliers$site[1]) |>
  summarize(n = n(), .by = c(genus, species, origin)) |>
  arrange(desc(n)) |>
  head(3) |>
  knitr::kable(
    caption = paste(
      "Top three most abundant taxa from",
      levels(arth$site_name)[as.numeric(outliers$site[1])],
      "outlier sample"
    )
  )
```

Table 1: Top three most abundant taxa from Laupāhoehoe outlier sample

genus	species	origin	n
Greenidea	formosana	adventive	423
Salina	celebensis	adventive	330
Entomobrya	socia	adventive	69

```
filter(arth, tree == outliers$tree[2] & site == outliers$site[2]) |>
  summarize(n = n(), .by = c(genus, species, origin)) |>
  arrange(desc(n)) |>
  head(3) |>
  knitr::kable(
    caption = paste(
      "Top three most abundant taxa from",
      levels(arth$site_name)[as.numeric(outliers$site[2])],
      "outlier sample"
    )
  )
```

Table 2: Top three most abundant taxa from Kohala outlier sample

genus	species	origin	n
Salina	celebensis	adventive	410
Forcipomyia	hardyi	endemic	92
Kilauea	sp_3	endemic	46

In both outliers the most abundant species is non-native, and in the case of Laupāhoehoe, all top three most abundant species are non-native. Laupāhoehoe and Kohala share *Salina celebensis* in common which is a Collembola known to reach high densities in Hawai'i [christiansen1992]. The abundances of *Salina celebensis* and *Greenidea formosana* in these samples are significant outliers compared to the maximum abundance found in any other sample as shown in Figure 2.

```
crazy_abund <- filter(arth,
  (site %in% outliers$site & tree %in% outliers$tree)) |>
  summarize(n = n(), .by = c(site, tree, species_code)) |>
  arrange(desc(n)) |>
  head(3)

summarize(arth, .by = c(site, tree),
  nmax = max(table(species_code))) |>
```



```

filter(nmax < 300) |>
ggplot(aes(nmax)) +
geom_histogram(bins = 20) +
geom_segment(data = crazy_abund,
             mapping = aes(x = n, y = rep(10, 3),
                           xend = n, yend = rep(0, 3)),
             color = "red",
             arrow = arrow(length = unit(1, "char")),
             inherit.aes = FALSE) +
xlab("Maximum abundances") +
ylab("Count") +
this_ms_look()

```

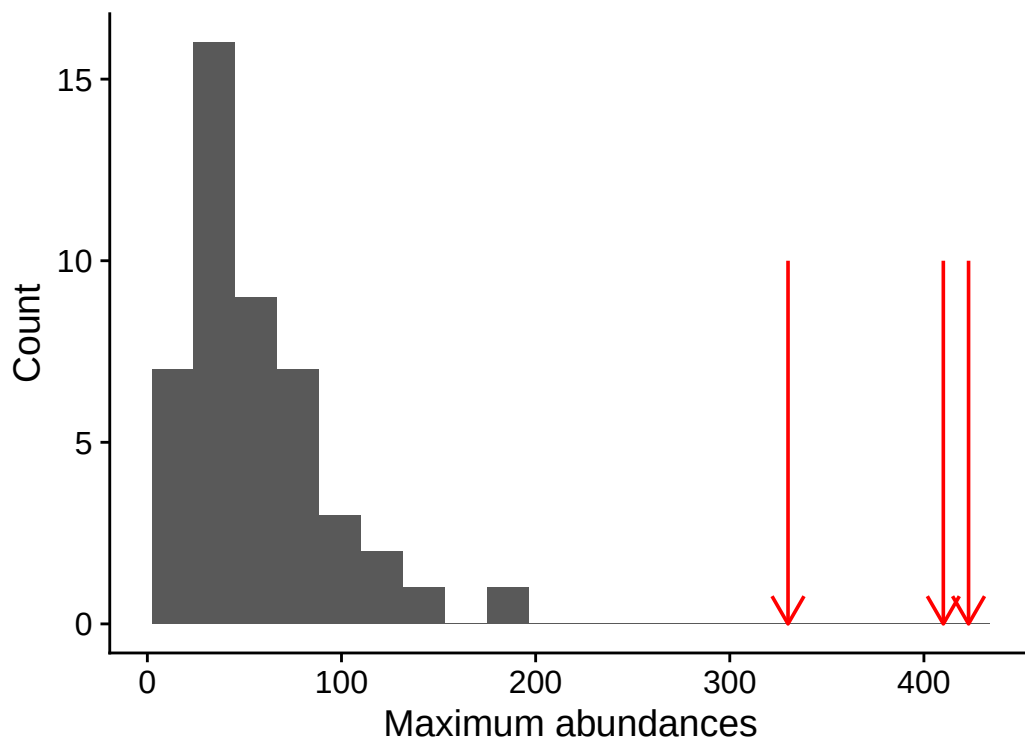


Figure 2

Because these samples are so unusual we decided to drop them from all further analyses presented in the main text. However, we retain the outlier samples in the supplement to validate that our results are not biased by their removal.

```

# remove from mete results
arth_mete_out <- arth_mete
arth_mete <- filter(arth_mete,
                    !((site %in% outliers$site) & (tree %in% outliers$tree)))

# remove from arth data frame
arth_out <- arth
arth <- filter(arth,
               !((site %in% outliers$site) & (tree %in% outliers$tree)))

```

Deviations from METE across the chronosequence

Now we analyze how deviations from METE behave across the chronosequence. The following code produces the analysis of a hump-shaped versus flat relationship between METE deviation and flow age.

```

# site-level means and sd for meta regression
site_summ <- summarize(arth_mete,
                       site_log_age = first(log(flow_age_my)),
                       sad_m = mean(log(z2_sad)),
                       sad_sd = sd(log(z2_sad)),
                       ipd_m = mean(log(z2_ipd)),
                       ipd_sd = sd(log(z2_ipd)),
                       n = n(),
                       .by = site)

# full model
sad_rma_chrono <- rma(yi = sad_m,
                     vi = sad_sd^2 / n,
                     mods = ~ site_log_age + I(site_log_age^2),
                     data = site_summ,
                     method = "FE")

# reduced model (for likelihood ratio test)
sad_rma_0 <- rma(yi = sad_m,
                 vi = sad_sd^2 / n,
                 mods = ~ 1,
                 data = site_summ,
                 method = "FE")

# full, reduced as with SAD

```

```

ipd_rma_chrono <- rma(yi = ipd_m,
                     vi = ipd_sd^2 / n,
                     mods = ~ site_log_age + I(site_log_age^2),
                     data = site_summ,
                     method = "FE")

ipd_rma_0 <- rma(yi = ipd_m,
                 vi = ipd_sd^2 / n,
                 mods = ~ 1,
                 data = site_summ,
                 method = "FE")

```

Likelihood ratio tests (LRT) can evaluate the significance of the quadratic model

```

# likelihood ratio test for sad across chrono
sad_lrt_chrono <- anova(sad_rma_0, sad_rma_chrono)
sad_lrt_chrono

```

	df	AIC	BIC	AICc	logLik	LRT	pval	QE	R ²
Full	3	9.2608	8.0891	33.2608	-1.6304			4.1970	
Reduced	1	26.9101	26.5196	28.2434	-12.4551	21.6493	<.0001	25.8463	67.5236%

```

ipd_lrt_chrono <- anova(ipd_rma_0, ipd_rma_chrono)
ipd_lrt_chrono

```

	df	AIC	BIC	AICc	logLik	LRT	pval	QE	R ²
Full	3	12.2792	11.1075	36.2792	-3.1396			0.3629	
Reduced	1	11.6584	11.2679	12.9918	-4.8292	3.3793	0.1846	3.7422	80.6036%

The quadratic model for SAD z^2 -values is supported by the LRT with a P -value < 0.0001, but the quadratic model for IPD z^2 -values did not have a significant P -value. However, the 95% CI for the quadratic term in the IPD z^2 -values model has the majority of its range below zero, indicating the data do tend toward a quadratic shape, just with greater variance compared to the quadratic model for SAD z^2 -values.

```

ipd_rma_chrono

```

Fixed-Effects with Moderators Model (k = 5)

I² (residual heterogeneity / unaccounted variability): 0.00%
H² (unaccounted variability / sampling variability): 0.18
R² (amount of heterogeneity accounted for): 80.60%

Test for Residual Heterogeneity:

QE(df = 2) = 0.3629, p-val = 0.8340

Test of Moderators (coefficients 2:3):

QM(df = 2) = 3.3793, p-val = 0.1846

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub
intrcpt	-0.0773	0.4466	-0.1732	0.8625	-0.9527	0.7980
site_log_age	-0.4558	0.2499	-1.8241	0.0681	-0.9456	0.0340 .
I(site_log_age ²)	-0.0609	0.0340	-1.7894	0.0736	-0.1276	0.0058 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The following code produces the visualization reported in the main text of deviation from METE across flow ages.

```
# prepare data frame of predicted values for plotting
x <- seq(min(site_summ$site_log_age),
        max(site_summ$site_log_age),
        length.out = 50)

newx <- cbind(1, x, x^2)
sad_hat <- predict(sad_rma_chrono, newmods = newx)
ipd_hat <- predict(ipd_rma_chrono, newmods = newx)

z2_chrono_pred <- rbind(
  data.frame(flow_age_my = exp(x),
             z2 = exp(sad_hat$pred),
             z2_lo = exp(sad_hat$ci.lb),
             z2_hi = exp(sad_hat$ci.ub),
             metric = "z2_sad"),
  data.frame(flow_age_my = exp(x),
```

```

        z2 = exp(ipd_hat$pred),
        z2_lo = exp(ipd_hat$ci.lb),
        z2_hi = exp(ipd_hat$ci.ub),
        metric = "z2_ipd")
)

# plotting
fig_z2_chrono <- pivot_longer(arth_mete, c(z2_sad, z2_ipd),
                              names_to = "metric", values_to = "z2") |>
  ggplot(aes(flow_age_my, z2, color = site_name)) +
  geom_jitter(width = 0.1) +
  # confidence envelope
  geom_ribbon(data = z2_chrono_pred,
            mapping = aes(x = flow_age_my,
                          ymin = z2_lo,
                          ymax = z2_hi),
            color = "gray", linewidth = 0.0001, alpha = 0.25) +
  # central tendency
  geom_line(data = z2_chrono_pred,
            mapping = aes(flow_age_my, z2),
            color = "black") +
  # facet by IPD or SAD and trick strip label into axis label
  facet_grid(
    rows = vars(metric),
    labeller = as_labeller(
      c(z2_ipd = "\"IPD\"~z^2*\"-value\"",
        z2_sad = "\"SAD\"~z^2*\"-value\""),
      label_parsed
    ),
    switch = "y"
  ) +
  this_ms_look(log_x = TRUE, log_y = TRUE) +
  theme(axis.title.y = element_blank(),
        strip.background = element_blank(),
        strip.placement = "outside",
        panel.border = element_rect(colour = "black",
                                     fill = NA,
                                     linewidth = 1))

fig_z2_chrono

```

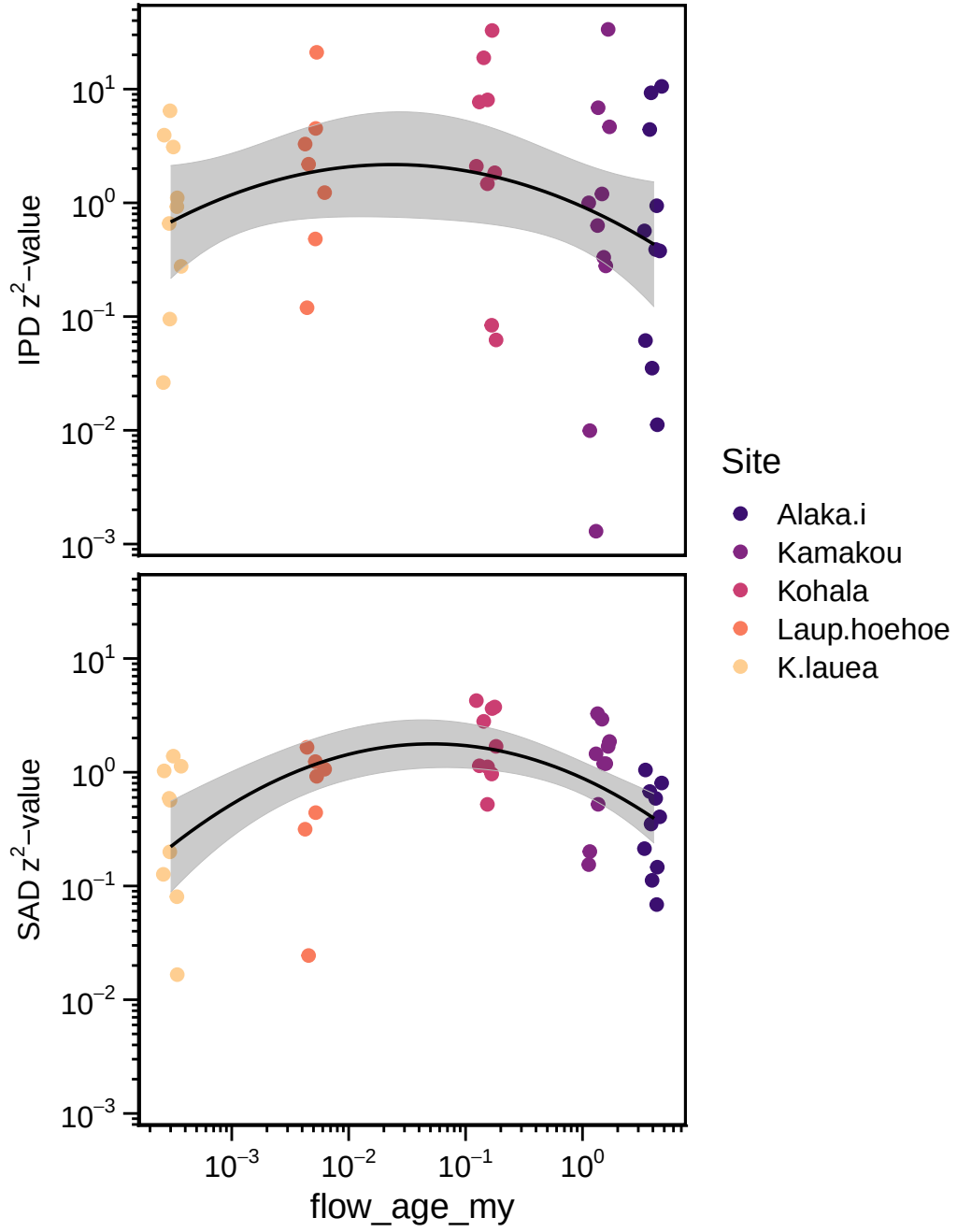
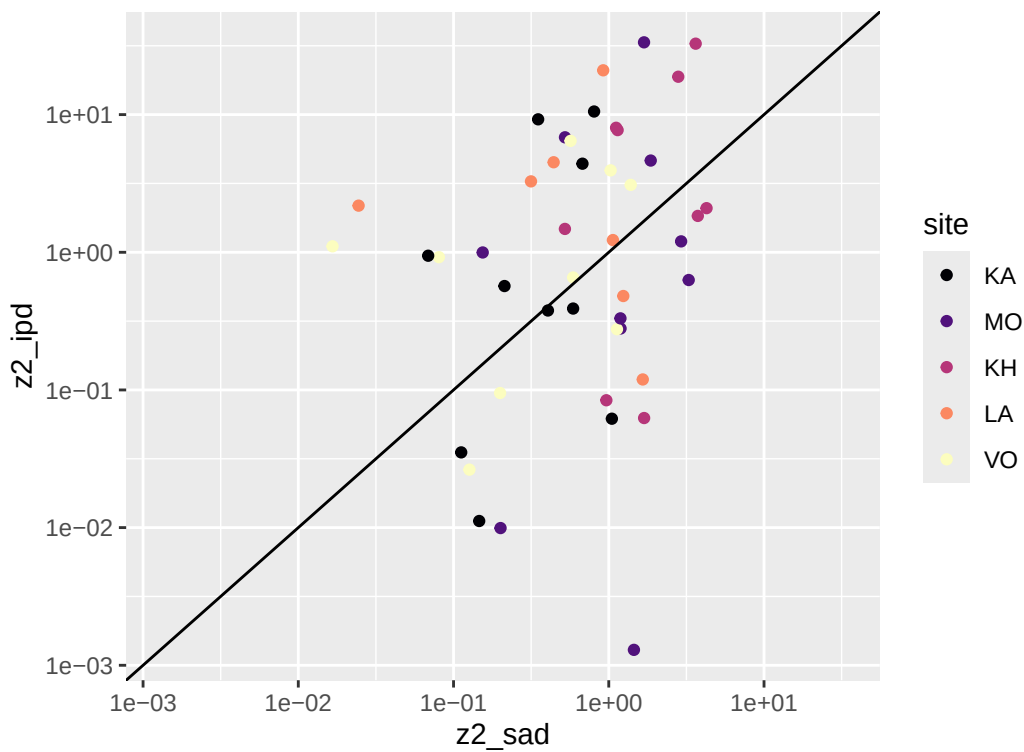


Figure 3: Deviations from METE as measured by z^2 across the chronosequence. Discussed in more detail in the main text.

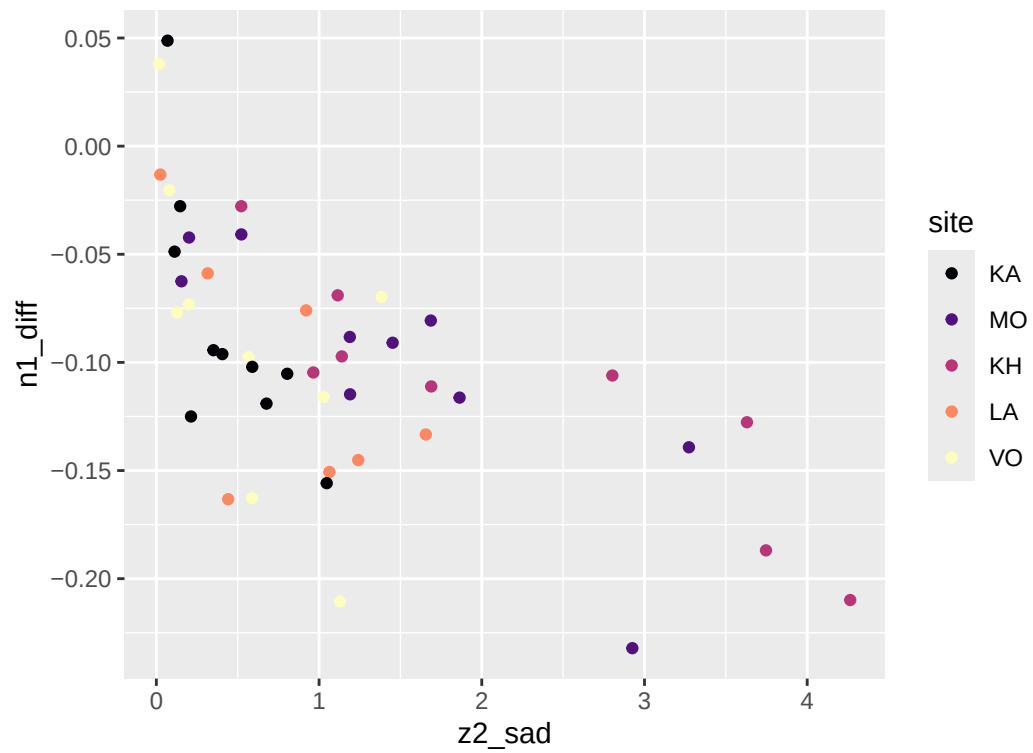
How do observed patterns deviate from METE?

?@fig-sad-how shows that when METE predictions deviate from observed SADs, METE under-predicts both rarity and dominance compared to observed SADs. This means that observed SADs are more uneven than the METE prediction.

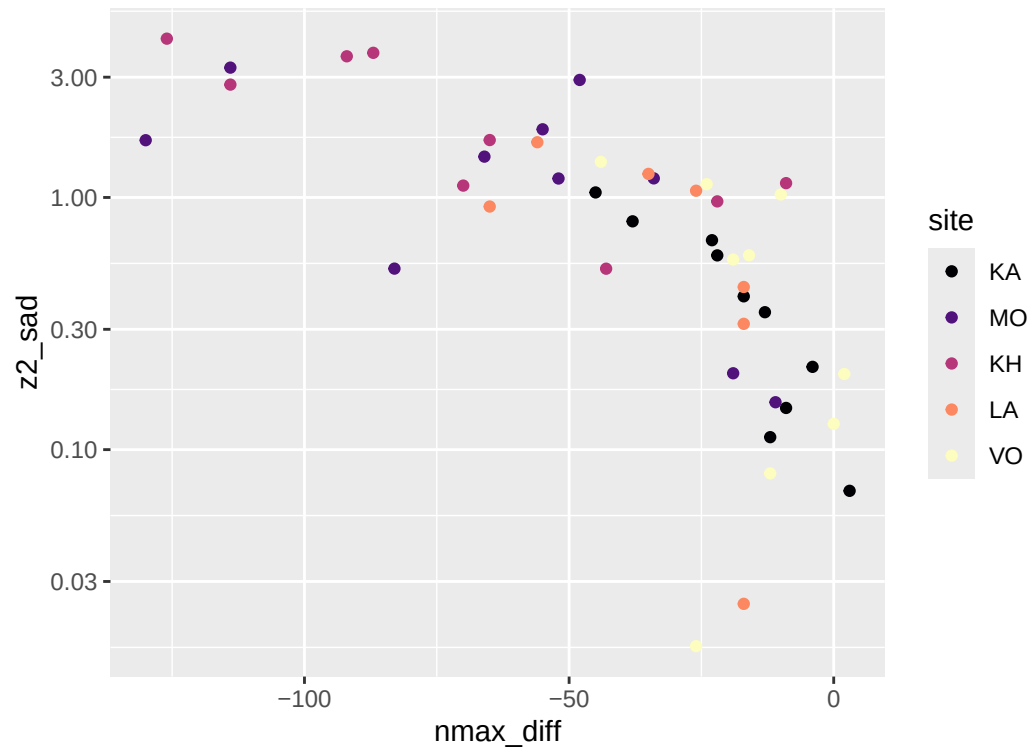
```
ggplot(arth_mete, aes(z2_sad, z2_ipd, color = site)) +  
  geom_point() +  
  scale_x_log10(limits = range(select(arth_mete, starts_with("z2")))) +  
  scale_y_log10(limits = range(select(arth_mete, starts_with("z2")))) +  
  scale_color_viridis_d(option = "magma") +  
  geom_abline(slope = 1, intercept = 0)
```



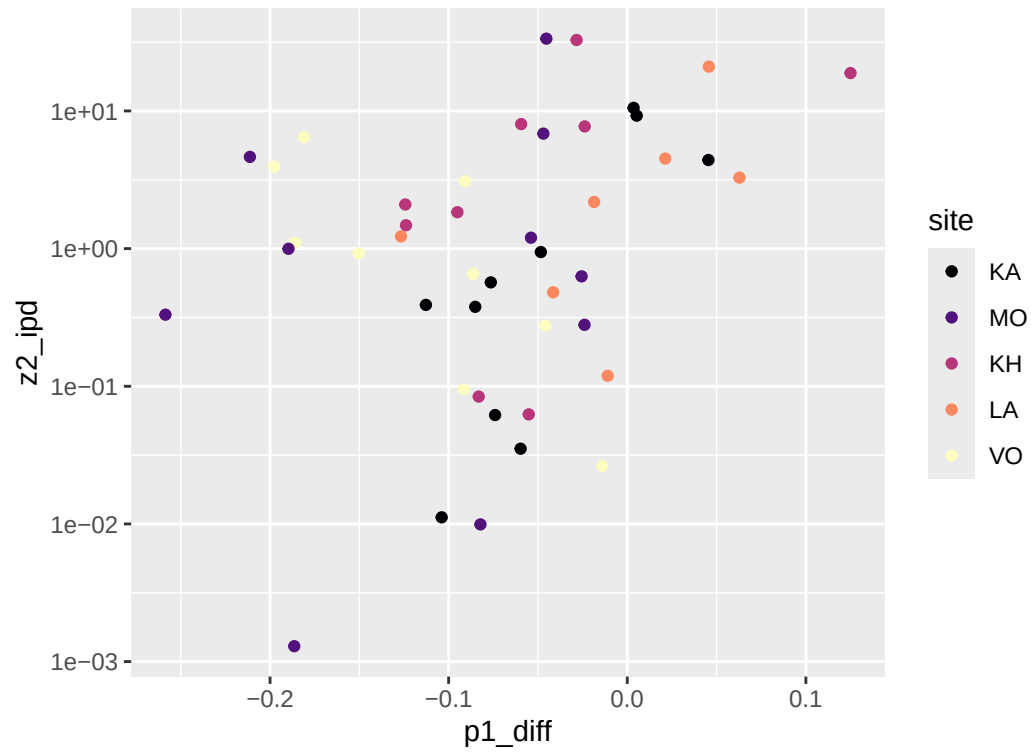
```
ggplot(arth_mete, aes(z2_sad, n1_diff, color = site)) +  
  geom_point() +  
  scale_color_viridis_d(option = "magma")
```



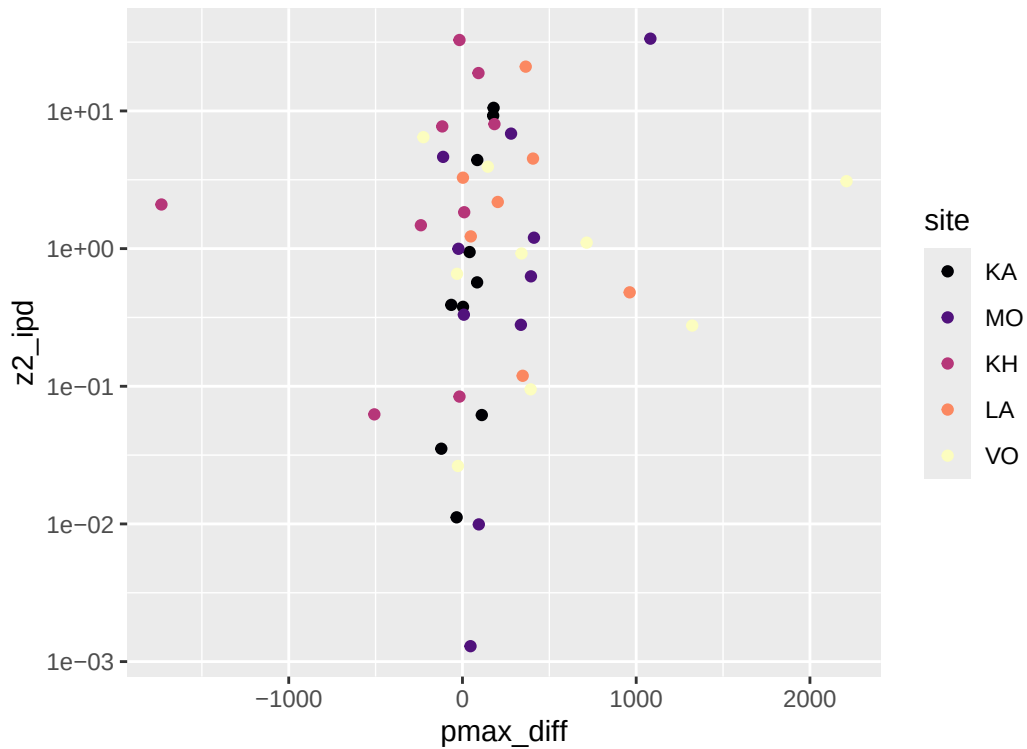
```
ggplot(arth_mete, aes(nmax_diff, z2_sad, color = site)) +  
  geom_point() +  
  scale_y_log10() +  
  scale_color_viridis_d(option = "magma")
```

```
ggplot(arth_mete, aes(p1_diff, z2_ipd, color = site)) +
  geom_point() +
  scale_y_log10() +
  scale_color_viridis_d(option = "magma")
```



```
ggplot(arth_mete, aes(pmax_diff, z2_ipd, color = site)) +  
  geom_point() +  
  scale_y_log10() +  
  scale_color_viridis_d(option = "magma")
```



How do METE and the real data differ

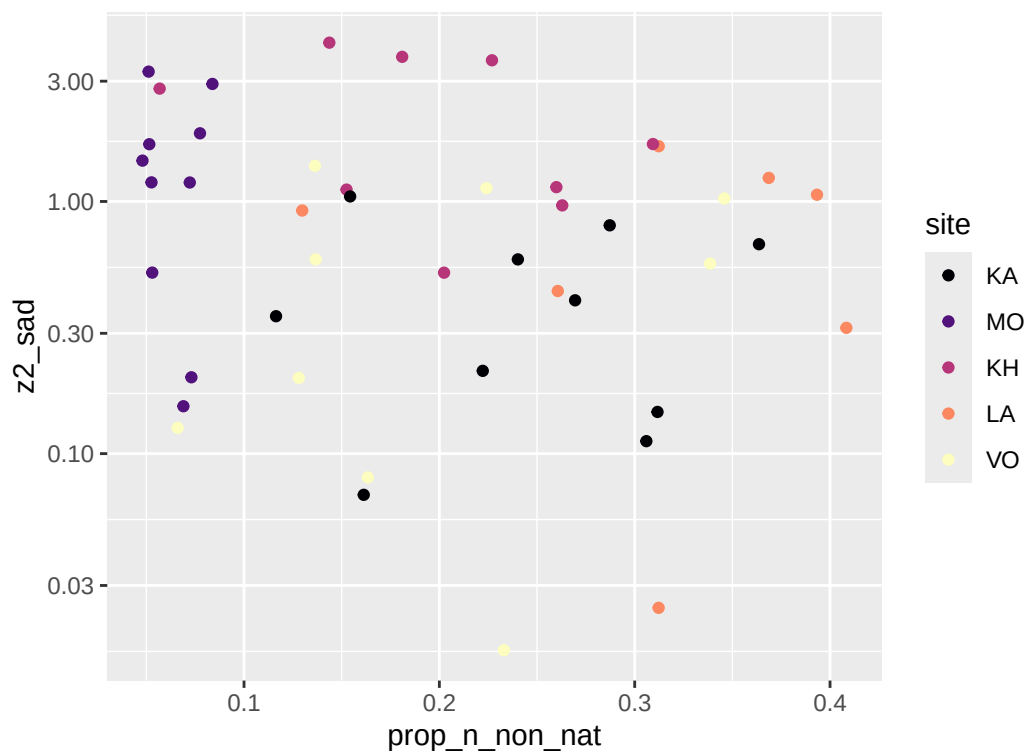
Which one over-predicts rarity?

Comparing deviations from METE with potential explanatory variables

Deviation vs. proportion that are invasive

```
ggplot(arth_mete, aes(prop_n_non_nat, z2_sad, color = site)) +
  geom_point() +
  scale_y_log10() +
  scale_color_viridis_d(option = "magma")

ggplot(arth_mete, aes(prop_p_non_nat, z2_ipd, color = site)) +
  geom_point() +
  scale_y_log10() +
  scale_color_viridis_d(option = "magma")
```



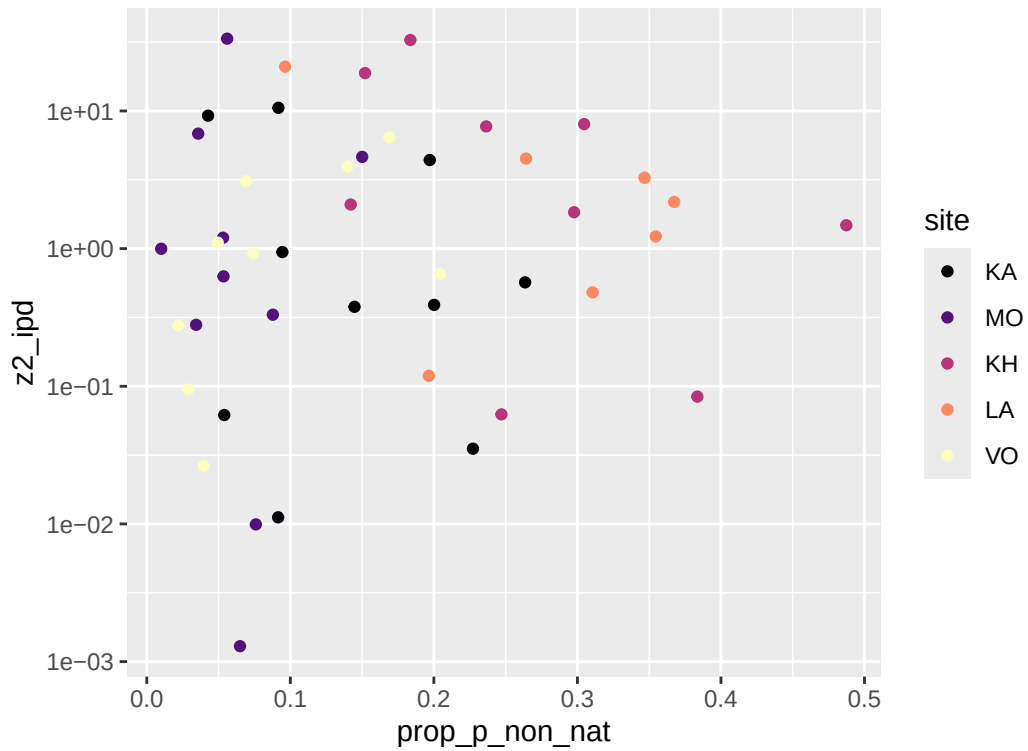


Figure 5

Deviation vs. dissimilarity

```
z_diss <- function(x, B = 500) {
  obs_dist <- x |> vegdist() |> mean()

  sim <- nullmodel(x, "quasiswap_count") |>
    simulate(nsim = B)

  sim_mean_dist <- sapply(1:dim(sim)[3], function(i) {
    vegdist(sim[, , i]) |> mean()
  })

  (obs_dist - mean(sim_mean_dist)) / sd(sim_mean_dist)
}
```

```

diss <- split(arth, arth$site) |>
  sapply(FUN = function(dat) {
    summarize(dat, abund = n(), .by = c(tree, species_code)) |>
      pivot_wider(names_from = species_code,
                  values_from = abund,
                  values_fill = 0) |>
      select(!tree) |>
      z_diss()
  })

mete_by_diss <- mutate(arth_mete,
                      mean_diss = diss[site])

ggplot(mete_by_diss, aes(mean_diss, z2_sad, color = site)) +
  geom_jitter(width = 0.25) +
  this_ms_look(log_y = TRUE)

xy <- mete_by_diss |>
  summarize(z2_sad_mean = mean(log(z2_sad)),
            z2_sad_sd = sd(log(z2_sad)),
            n = n(),
            mean_diss = first(mean_diss),
            .by = site)

rma(yi = z2_sad_mean,
    vi = z2_sad_sd^2 / n,
    mods = ~ mean_diss,
    data = xy,
    method = "REML")

```